5 Fig. 5.1 shows a heater in a bathroom.

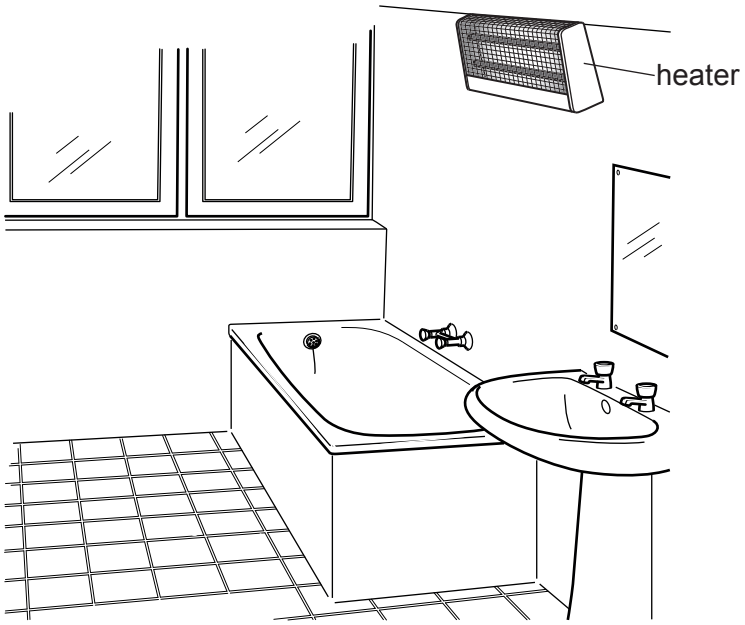


Fig. 5.1

The heater is at a very high temperature and it glows red. The manufacturer states:

> "The heater emits light and radiation and it transfers thermal energy by radiation."

(a) State the part of the electromagnetic spectrum that transfers thermal energy.

..... [1]

(b) State:

(i) **one** way in which visible light and the radiation identified in **(a)** are similar

.....

..... [1]

(ii) **one** way in which visible light differs from the radiation identified in **(a)**.

.....

..... [1]

(c) Some surfaces are better at emitting radiation than others.

(i) Describe an experiment to show whether a black surface or a white surface is the better emitter of radiation. You may draw a diagram.

.....

.....

.....

.....

..... [3]

(ii) To ensure that the conclusion reached in the experiment in **(c)(i)** is correct, several details of the experiment must be identical when testing the two different surfaces.

State **two** quantities in the experiment that you described that must be identical during the test.

1. .....

.....

2. .....

.....

[2]

[Total: 8]